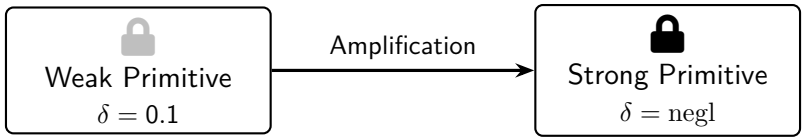
Optimal Amplification via Bias-Resilient Combiners

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Security Amplification

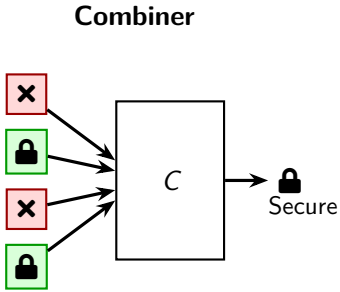


The Goal: Transform weakly secure primitive into strongly secure.

- **Simplicity:** Easier to construct weak primitives.
- **Robustness:** Tolerate underlying weakness.

OWF [Yao82], PKE [DNR04, HR05, LT13], KA [Hol05], OT [CK90, DKS99, Wul07], NIZK [GJS19, BKP+24, BG24, AK25], and many others.

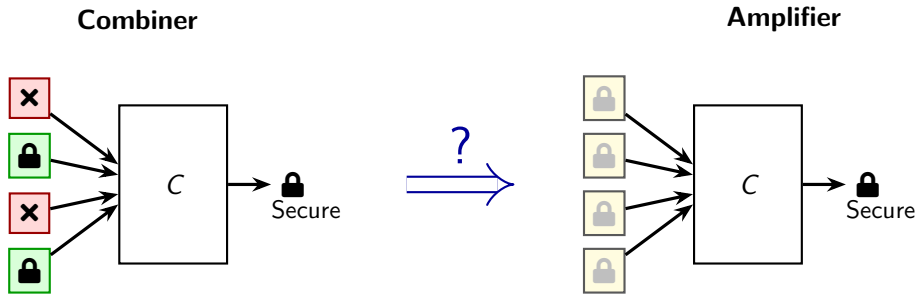
Combiners and Amplifiers



$\geq k$ secure $\Rightarrow$ construction secure

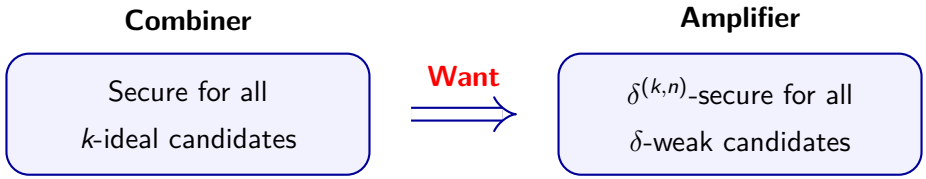
Robust Combiner: Guarantees security if k out of n candidates secure.
[Harnik–Kilian–Naor–Reingold–Rosen05, Herzberg05]

Combiners and Amplifiers



Do combiners inherently amplify?

Idealized Combiner-is-Amplifier Theorem

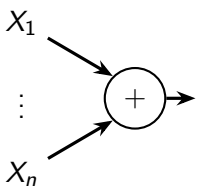


$$\delta^{(k,n)} := \Pr[\mathrm{Bin}(n, 1-\delta) < k]$$

$$= \sum_{j=0}^{k-1} \binom{n}{j} (1-\delta)^j \delta^{n-j}.$$

All-or-nothing intuition: each candidate ideal w.p. $1 - \delta$.

The XOR Lemma



Combiner? Yes, 1-out-of-n.

Amplifier? (any nontrivial δ)

Bits	Yao's XOR lemma [Yao82]
Bit strings ($\mathbb{F}_{2^m}$)	No! Next slide
Prime fields ($\mathbb{F}_p$)	Yes! Later

The Barrier: Consistent Small Bias

Idealized theorem false

Bitwise XOR: $X_1 \oplus \cdots \oplus X_n$

Uniform inputs with LSB 0: $\delta = 1/2$.

XOR always has LSB 0.

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Bypassing the Barrier

Quantitative Restriction [Applebaum–Bitansky–G26]

Any combiner

Restrict $\delta < 1/2$

Quantify max penalty

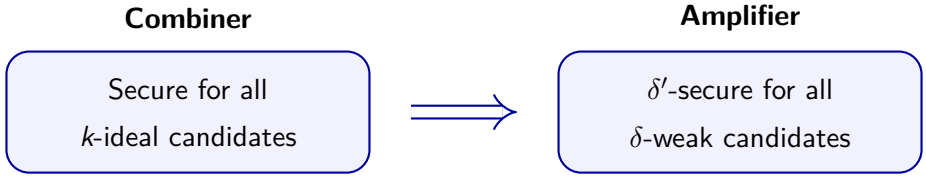
Qualitative Restriction (this talk)

Strengthen combiner

Any $\delta < 1$

No penalty

Prior Work: Amplification Penalty [ABG26]



$$\delta' := \binom{n}{k-1}(2\delta)^{n-k+1}.$$

- Inherent restriction $\delta < 1/2$.
- Exponential penalty: $\delta' \geq 2^{n-k+1}\,\delta^{(k,n)}$.

Related work: [Maurer–Pietrzak–Renner07, Maurer–Tessaro09, Lanzenberger–Maurer20]

Optimal Amplification via Bias-Resilience

Bias-resilient combiner

Secure for all

k-almost-ideal candidates

$\Longrightarrow$

Amplifier

$\delta^{(k,n)}$-secure for all

δ-weak candidates

Relax "good candidate"

$$\underbrace{S_i = I_i}_{\text{Ideal}} \quad \longrightarrow \quad \underbrace{\max_x \frac{S_i(x)}{I_i(x)} \leq \frac{1}{1-\delta}}_{\text{Almost-ideal}}$$

Prime-Field XOR: Bias-Resilience Perspective

Independent $X_1, \ldots, X_n \in \mathbb{F}_p$:

$$\forall i: \; X_i \approx_{1/2} U_p \qquad \textbf{Goal:} \quad \sum_{i=1}^{n} X_i \bmod p \approx_\varepsilon U_p.$$

Unpredictability [Shimizu–Yasunaga26]

All n copies

≈ 1 bit of min-entropy per copy

Bias-resilience

$\approx n/2$ good components

$\log p - 1$ bits of min-entropy per good component

The Prime-Field XOR Lemma

$$\forall i:\ X_i \approx_\delta U_p \qquad \Longrightarrow \qquad \sum_{i=1}^{n} X_i \bmod p \approx_\varepsilon U_p.$$

Repetition bound

$$\gamma := (1 - 1/p) - \delta > 0, \qquad n \approx \frac{\log(1/\varepsilon)}{\gamma^3}.$$

Also holds computationally (hardcore lemma).

Fourier analysis builds on [Benhamouda–Degwekar–Ishai–Rabin18].

Example: $\delta = 1/2$

$$\underbrace{\log(1/\varepsilon)}_{\text{Bias-resilience}} \quad \text{vs.} \quad \underbrace{p^2 \log(p/\varepsilon)}_{\text{Unpredictability}} \quad \text{copies.}$$

All-or-Nothing Amplification Penalty

All-or-nothing amplifier: analyzed in weaker attack model.

Each candidate ideal w.p. ρ.

Corresponding security error: ε.

Standard weakness: $R_i \approx_\delta I_i$.

More attacks. Bound new security error ε'.

Multiplicative penalty bound

$$\varepsilon' \leq \varepsilon \cdot \underbrace{\left(1 + \frac{2\rho\delta}{1-\rho}\right)^n}_{\text{tight penalty}}.$$

Thank You!

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